

Day 17 Plan

Maths: Trigonometry, Heights & Distances · Reasoning: Figure Counting, Dice & Cubes · Science: Biology 4: Plants, Heredity, Evolution & Ecology

Today's schedule (about 5½ hours with breaks)

#	Block	Time	What you do	Where
0	Day 16 recall	15 min	10 questions from Day 16, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	5 simplification questions for speed → formulas & methods (10 min) → 2-3 book questions per pattern, timed → 10 Testbook PYQs on this topic → log mistakes	Section A · book · Testbook
-	Break	10 min	Away from the screen	-
2	Reasoning	45 min	Learn the method (10 min) → 15 timed questions, Testbook PYQs first → log	Section B · Testbook
-	Break	10 min	Away from the screen	-
3	Science	110 min	7 lessons with quick checks (≈65 min) → 20-question practice set → Testbook PYQs: plant biology, heredity, environment (A/B tier)	Section C · Testbook
-	Break	10 min	Away from the screen	-
4	General awareness	45 min	20 current-affairs MCQs (20 min) → static GA: Ancient & medieval history — famous facts only (B tier) — Testbook GA PYQs on it (25 min)	CA PDF · Testbook
5	Close	15 min	Redo today's wrong questions · write 3 rules for tomorrow · send me your scores	Mistake log

Short day? Cut in this order: Testbook science PYQs → reasoning down to 10 questions → static GA down to 10 min. Never cut recall, maths or the science lessons. · Exam pace (average 54 sec a question): GA ~30 sec · Science ~45 · Reasoning ~50 · Maths ~65.

Block 0 — Day 16 recall (answers on the last page)

#	Question	Your answer
1	Universal donor blood group?	
2	Lifespan of a red blood cell?	
3	Pulmonary artery carries?	
4	Functional unit of the kidney?	
5	Part of the brain for balance?	
6	Master gland?	
7	Smallest bone?	
8	Product of anaerobic respiration in muscles?	
9	Diagonals in a hexagon?	
10	Mirror image of a clock showing 4:20?	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — Trigonometry, Heights & Distances

Core idea

Trigonometry in RRB papers is the **standard-values table**, three identities and the complementary-angle rules. Heights and distances is just **$\tan \theta = \text{height} \div \text{distance}$** on a right-angled triangle. Learn the table until you can write it in 30 seconds.

Formulas

Concept	Formula / rule
Ratios	$\sin = \text{opposite/hypotenuse} \cdot \cos = \text{adjacent/hypotenuse} \cdot \tan = \text{opposite/adjacent} \cdot \text{cosec, sec, cot are reciprocals}$
$\sin 0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$	0, $\frac{1}{2}$, $\frac{1}{\sqrt{2}}$, $\frac{\sqrt{3}}{2}$, 1
$\cos 0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$	1, $\frac{\sqrt{3}}{2}$, $\frac{1}{\sqrt{2}}$, $\frac{1}{2}$, 0
$\tan 0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$	0, $\frac{1}{\sqrt{3}}$, 1, $\sqrt{3}$, not defined
Identities	$\sin^2\theta + \cos^2\theta = 1 \cdot 1 + \tan^2\theta = \sec^2\theta \cdot 1 + \cot^2\theta = \text{cosec}^2\theta$
Complementary angles	$\sin(90^\circ - \theta) = \cos \theta \cdot \tan(90^\circ - \theta) = \cot \theta \cdot \sec(90^\circ - \theta) = \text{cosec } \theta$
Heights & distances	$\tan \theta = \text{height} \div \text{horizontal distance} \cdot \text{angle of elevation} = \text{angle of depression}$
Radian	$\pi \text{ radians} = 180^\circ$

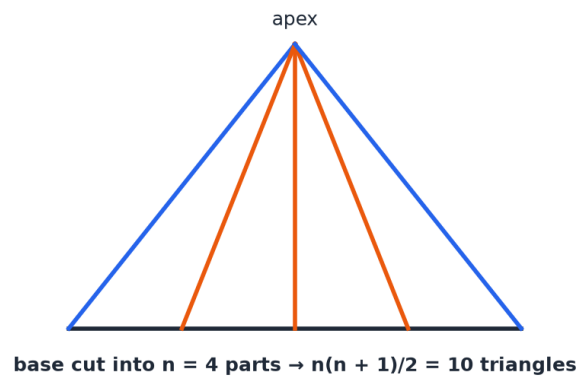
Patterns — the method and one solved example each

Pattern	Method	Solved example
1. Values	Table	$\sin 30^\circ + \cos 60^\circ = \frac{1}{2} + \frac{1}{2} = \mathbf{1}$
2. Values squared	Table	$\tan^2 60^\circ + \sin^2 45^\circ = 3 + \frac{1}{2} = \mathbf{3.5}$
3. One ratio given	3-4-5 triangle	$\sin \theta = \frac{3}{5} \rightarrow \cos \theta = \mathbf{\frac{4}{5}}$, $\tan \theta = \mathbf{\frac{3}{4}}$
4. Complementary	$\sin^2 + \cos^2 = 1$	$\sin^2 37^\circ + \sin^2 53^\circ = \sin^2 37^\circ + \cos^2 37^\circ = \mathbf{1}$
5. Complementary product	$\tan \times \cot$	$\tan 10^\circ \times \tan 80^\circ = \tan 10^\circ \times \cot 10^\circ = \mathbf{1}$
6. Identity	$\sec^2 - \tan^2 = 1$	$\sec^2 \theta - \tan^2 \theta = \mathbf{1}$
7. Tower	$h = d \tan \theta$	Elevation 60° from 20 m away $\rightarrow h = 20\sqrt{3} \approx \mathbf{34.64 \text{ m}}$
8. Ladder	$h = L \sin \theta$	10 m ladder at 60° to the ground $\rightarrow$ reaches $5\sqrt{3} \approx \mathbf{8.66 \text{ m}}$
9. Shadow	$s = h \div \tan \theta$	10 m pole, Sun at $45^\circ \rightarrow$ shadow $\mathbf{10 \text{ m}}$; at $30^\circ \rightarrow \mathbf{10\sqrt{3} \text{ m}}$
10. Two angles	Two equations	Elevation 30° then 60° after walking 20 m closer $\rightarrow x\sqrt{3} = (x + 20)/\sqrt{3} \rightarrow x = 10 \rightarrow h = \mathbf{10\sqrt{3} \approx 17.32 \text{ m}}$

Traps

- $\tan 30^\circ = 1/\sqrt{3}$ and $\tan 60^\circ = \sqrt{3}$ — don't swap them.
- Angles are measured from the **horizontal**.
- The ladder is the hypotenuse: use sin for height, cos for the distance from the wall.

B · REASONING — Figure Counting, Dice & Cubes



Count	Rule
Triangles: lines from the apex cut the base into n parts	$n(n + 1)/2$
Squares in an $n \times n$ grid	$1^2 + 2^2 + \dots + n^2$ (8×8 chessboard $\rightarrow 204$)
Rectangles in an $m \times n$ grid	$[m(m + 1)/2] \times [n(n + 1)/2]$ (squares included)

Dice: on a standard die, opposite faces add to **7** (1-6, 2-5, 3-4). Two faces that appear side by side are **never** opposite. If one face is common to two views, the four faces seen next to it are its neighbours, so the face left over is opposite it.

Cube $n \times n \times n$, painted outside, cut into unit cubes	How many
3 faces painted (corners)	8
2 faces painted (edges)	$12(n - 2)$
1 face painted (face centres)	$6(n - 2)^2$
No face painted (inside)	$(n - 2)^3$

Patterns, with solved examples

Pattern	Example	Solution
Triangles	3 lines from the apex divide the base into 4 parts. How many triangles?	$4 \times 5 \div 2 = \mathbf{10}$
Squares	How many squares in a 4×4 grid?	$1 + 4 + 9 + 16 = \mathbf{30}$
Painted cube	A $4 \times 4 \times 4$ cube is painted and cut into 64 small cubes. How many have 2 faces painted? None?	$12 \times 2 = \mathbf{24} \cdot 2^3 = \mathbf{8}$
Standard die	On a standard die, which number is opposite 4?	$7 - 4 = \mathbf{3}$
Two views	Two views of a die both show 1; next to it you see 2, 3 in one and 4, 5 in the other. Opposite 1?	Neighbours 2, 3, 4, 5 $\rightarrow \mathbf{6}$

Speed rule

For figure counting, count systematically by size (single pieces, then pairs, then threes...) — random counting always misses some.

C · SCIENCE — Biology 4: Plants, Heredity, Evolution & Ecology

7 lessons that finish biology. Environment and ecology were 9 of 210 (R); plant biology, heredity and evolution are B tier — know the facts in the tables and move on.

Lesson 1 · Photosynthesis and modes of nutrition (about 9 min)

$6\text{CO}_2 + 12\text{H}_2\text{O} \rightarrow (\text{sunlight, chlorophyll}) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 + 6\text{H}_2\text{O}$. It happens in **chloroplasts**. CO_2 enters through **stomata**; water comes up from the roots. The **oxygen released comes from water**, not from CO_2 . Photosynthesis stores energy — it is **endothermic**. Test a leaf for starch with iodine (blue-black).

Mode	Meaning	Example
Autotrophic	Makes its own food	Green plants
Saprophytic	Feeds on dead, decaying matter	Fungi (mushroom, bread mould)
Parasitic	Lives on a host and harms it	Cuscuta (amarbel), tapeworm
Insectivorous	Traps insects for nitrogen	Pitcher plant , Venus flytrap
Symbiotic	Two organisms help each other	Lichen (alga + fungus), Rhizobium in legumes

Quick check 1 — answer out loud before moving on (answers at the end)

a) Where does the oxygen given off in photosynthesis come from? b) Name a parasitic plant.

Lesson 2 · Transport in plants (about 6 min)

Xylem carries water and minerals upward; the main pulling force is **transpiration** — the loss of water vapour through stomata, which also cools the plant. **Phloem** carries food (sugar) in both directions — **translocation**. Root hairs take in water by osmosis. Stomata open and close using **guard cells**; woody stems also exchange gases through lenticels.

Quick check 2 — answer out loud before moving on (answers at the end)

What is the loss of water vapour from leaves called?

Lesson 3 · Plant hormones and movements (about 8 min)

Hormone	Effect	
Auxin	Cell elongation at shoot tips; makes shoots bend towards light	
Gibberellin	Stem elongation; bolting; seed germination	
Cytokinin	Cell division; delays ageing of leaves	
Abscisic acid (ABA)	Growth inhibitor ; closes stomata; 'stress hormone'; leaf fall	
Ethylene	Gaseous ; ripens fruit	
Movement	Response to	Shoot / root
Phototropism	Light	Shoot towards (+), root away (–)
Geotropism	Gravity	Root towards (+), shoot away (–)

Movement	Response to	Shoot / root
Hydrotropism · chemotropism · thigmotropism	Water · chemicals · touch	Roots to water · pollen tube to ovule · tendrils coil

Touch-me-not (*Mimosa pudica*) folds its leaves on touch — a **nastic** movement, not growth. **J. C. Bose** invented the crescograph and showed that plants respond to stimuli.

Quick check 3 — answer out loud before moving on (answers at the end)

a) Gaseous hormone that ripens fruit? b) Which hormone is a growth inhibitor?

Lesson 4 · Reproduction in plants and simple organisms (about 8 min)

Flower parts: sepals, petals, **stamen** (male: anther + filament, makes pollen) and **pistil/carpel** (female: stigma, style, ovary). After pollination (by wind, water or insects) and fertilisation, the **ovary becomes the fruit** and the **ovule becomes the seed**.

Method	Example
Stem tuber ('eyes')	Potato (a modified stem)
Rhizome	Ginger, turmeric
Bulb	Onion
Leaf buds	Bryophyllum
Root	Sweet potato, dahlia
Grafting · layering	Mango, rose · jasmine
Binary fission	Amoeba, bacteria
Budding	Yeast, Hydra
Spores	Rhizopus (bread mould), ferns
Regeneration	Planaria

Quick check 4 — answer out loud before moving on (answers at the end)

a) The potato is which part of the plant? b) How does Bryophyllum reproduce vegetatively?

Lesson 5 · Heredity (about 9 min)

Gregor Mendel (father of genetics) worked on **pea plants**. Each trait is controlled by a pair of factors (genes); one can be **dominant** (T, tall) and hide the **recessive** one (t, dwarf).

Tt × Tt	T	t
T	TT (tall)	Tt (tall)
t	Tt (tall)	tt (dwarf)

Monohybrid cross F₂: phenotype **3 tall : 1 dwarf**; genotype **1 TT : 2 Tt : 1 tt**. Dihybrid F₂ phenotype: **9 : 3 : 3 : 1**.

Disorder	Cause
Colour blindness, haemophilia	Genes on the X chromosome (more common in males)
Down syndrome	An extra chromosome 21 (trisomy 21)
Sickle-cell anaemia	Faulty haemoglobin gene

Quick check 5 — answer out loud before moving on (answers at the end)

a) Phenotypic ratio in a monohybrid F₂ generation? b) Down syndrome is caused by?

Lesson 6 · Evolution (about 7 min)

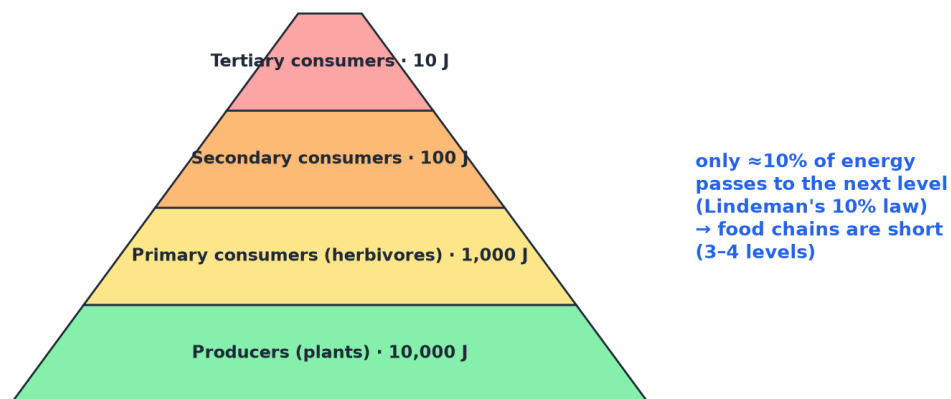
Idea	Who / what
Natural selection	Charles Darwin — 'On the Origin of Species' (1859); survival of the fittest
Inheritance of acquired characters	Lamarck — use and disuse (the giraffe's neck); now rejected
Homologous organs	Same origin, different jobs — human arm, bat wing, whale flipper
Analogous organs	Different origin, same job — wings of a bird and an insect
Vestigial organs (humans)	Appendix, wisdom teeth, tail bone (coccyx), ear muscles
Fossils	Age found by carbon-14 dating

Quick check 6 — answer out loud before moving on (answers at the end)

a) Are the wings of a bird and an insect homologous or analogous? b) Who wrote 'On the Origin of Species'?

Lesson 7 · Ecology and environment (about 10 min)

An **ecosystem** has biotic parts (producers, consumers, decomposers) and abiotic parts (air, water, soil, light). **Food chain:** grass → deer → lion. Decomposers (bacteria, fungi) return nutrients to the soil.



Biomagnification: non-biodegradable chemicals such as **DDT** build up at each level, so top predators have the most.

Problem	Cause	Note
Ozone depletion	CFCs (fridges, ACs, aerosols)	Ozone layer is in the stratosphere and blocks UV · Montreal Protocol, 1987
Global warming	Greenhouse gases: CO ₂ , methane, N ₂ O, CFCs, water vapour	Kyoto Protocol (1997), Paris Agreement (2015)
Acid rain	SO ₂ and NO _x	pH below 5.6; damages marble (Taj Mahal)
Eutrophication	Fertiliser run-off into water	Algal bloom uses up oxygen; fish die

Quick check 7 — answer out loud before moving on (answers at the end)

a) What fraction of energy passes from one level of a food chain to the next? b) Which chemicals deplete the ozone layer?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Photosynthesis	Chloroplast · O ₂ comes from water · endothermic · Cuscuta parasite · pitcher plant insectivorous · lichen symbiotic
Transport	Xylem water up (transpiration pull) · phloem food both ways (translocation)
Hormones	Auxin elongation, phototropism · gibberellin stem · cytokinin division · ABA inhibitor · ethylene gas, ripening
Reproduction	Ovary → fruit, ovule → seed · potato stem tuber · Bryophyllum leaf buds · yeast budding · Amoeba fission
Heredity	Mendel peas · 3 : 1, 1 : 2 : 1, 9 : 3 : 3 : 1 · Down syndrome trisomy 21 · haemophilia, colour blindness X-linked
Evolution	Darwin natural selection (1859) · Lamarck acquired characters · homologous same origin · analogous same job · appendix vestigial
Ecology	10% law · DDT biomagnification · CFC → ozone (Montreal 1987) · acid rain SO ₂ , NO _x , pH < 5.6

Science practice set — 20 questions · target 6 minutes

#	Question
1	The oxygen released in photosynthesis comes from: (a) carbon dioxide (b) glucose (c) chlorophyll (d) water
2	Photosynthesis takes place in the: (a) mitochondrion (b) nucleus (c) ribosome (d) chloroplast
3	Cuscuta (amarbel) is a: (a) saprophyte (b) parasite (c) insectivorous plant (d) symbiont
4	A lichen is an association of: (a) an alga and a fungus (b) a bacterium and a virus (c) two fungi (d) a moss and a fern
5	Loss of water vapour from the aerial parts of a plant is called: (a) translocation (b) guttation (c) transpiration (d) respiration
6	Food is carried in plants by: (a) xylem (b) root hairs (c) phloem (d) stomata
7	The plant hormone that ripens fruit is: (a) ethylene (b) auxin (c) cytokinin (d) gibberellin
8	Abscisic acid is a: (a) growth promoter (b) fruit-ripening gas (c) growth inhibitor (d) cell-division hormone
9	Roots growing towards gravity is called: (a) negative geotropism (b) phototropism (c) positive geotropism (d) thigmotropism
10	The crescograph was invented by: (a) J. C. Bose (b) C. V. Raman (c) Gregor Mendel (d) Charles Darwin
11	After fertilisation, the ovary develops into the: (a) seed (b) flower (c) fruit (d) embryo only
12	The potato is a modified: (a) root (b) leaf (c) fruit (d) stem
13	Yeast reproduces asexually by: (a) binary fission (b) spores only (c) fragmentation (d) budding
14	Mendel did his experiments on: (a) fruit flies (b) maize (c) bacteria (d) pea plants
15	The phenotypic ratio of a monohybrid F ₂ generation is: (a) 1 : 2 : 1 (b) 9 : 3 : 3 : 1 (c) 3 : 1 (d) 1 : 1
16	Down syndrome is caused by: (a) a missing X chromosome (b) a vitamin deficiency (c) an extra chromosome 21 (d) a virus
17	The wing of a bat and the arm of a human are: (a) analogous organs (b) vestigial organs (c) unrelated (d) homologous organs
18	'On the Origin of Species' was written by: (a) Charles Darwin (b) Lamarck (c) Mendel (d) Hugo de Vries
19	About how much energy passes to the next level in a food chain? (a) 50% (b) 1% (c) 90% (d) 10%
20	The ozone layer is mainly depleted by: (a) CFCs (b) carbon dioxide (c) nitrogen (d) oxygen

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	d	Lesson 1
2	d	Lesson 1
3	b	Lesson 1
4	a	Lesson 1
5	c	Lesson 2
6	c	Lesson 2
7	a	Lesson 3
8	c	Lesson 3
9	c	Lesson 3
10	a	Lesson 3
11	c	Lesson 4
12	d	Lesson 4
13	d	Lesson 4
14	d	Lesson 5
15	c	Lesson 5
16	c	Lesson 5
17	d	Lesson 6
18	a	Lesson 6
19	d	Lesson 7
20	a	Lesson 7

Quick checks

QC	Answers
1	a) From water (split during the light reaction) b) Cuscuta (amarbel)
2	Transpiration
3	a) Ethylene b) Absciscic acid
4	a) The stem (a stem tuber) b) Through buds on the edges of its leaves
5	a) 3 : 1 b) An extra copy of chromosome 21 (trisomy 21)
6	a) Analogous b) Charles Darwin
7	a) About 10% b) CFCs (chlorofluorocarbons)

Day 16 recall (Block 0)

#	Answer	#	Answer
1	O negative	6	Pituitary
2	≈ 120 days	7	Stapes
3	Deoxygenated blood	8	Lactic acid
4	Nephron	9	9
5	Cerebellum	10	7:40

Send me tonight

Day 16 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.